

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SESSIONAL TEST -2nd (SET-2)

Course: B.Tech.

Session: 2025-26

Time: 1 hr.

Sem: 1st

Subject Code: BCS-101

Sec: E, F, G and H

Subject: Programming For Problem Solving

Max. Marks: 20

Q.No. :	1	2	3	4	5	6
CO No. :	4	4	4	5	5	3
Bloom's Level	K3	K2,K3	K1,K3	K2	K2	K3

CO3: To implement conditional branching, iteration and recursion.

CO4: To decompose a problem into functions and synthesize a complete program using divide and conquer approach.

CO5: To use arrays, pointers and structures to develop algorithms and programs.

Note:

1. This paper contains two sections (A) and (B)
2. All sections are compulsory.

SECTION- A

1.	Write a program to search an element in an array using linear search method.	2 marks
2.	Write a program to sort elements of the 1D array using selection sort method and explain it with an example.	3 marks
3.	Define function and write a program to swap two numbers using call by value and call by reference method.	3 marks

SECTION- B

4.	Discuss dynamic memory allocation in details. Also explain malloc(), calloc(), realloc() and free() functions with syntax.	4 marks
5.	Explain Macro and its types with suitable example of each in details.	4 marks
6.	Write a program to multiply two matrices of nxn order.	4 marks

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SESSIONAL TEST -2nd (SET-2)

Course: B.Tech.

Sem: 2nd

Session: 2024-25

Subject: Programming For Problem Solving

Subject Code: BCS-201

Time: 1 hr.

See A, B, C and D

Give marks to

Q.No. :	1	2	3	4	5	6
CO No. :	3	3	4	3	5	6
Bloom's Level	K3	K2	K3	K3	K2	K2

CO3: To implement conditional branching, iteration and recursion.

CO4: To decompose a problem into functions and synthesize a complete program using top-down and conquer approach.

CO5: To use arrays, pointers and structures to develop algorithms and programs.

Note:

1. This paper contains two sections (A) and (B)

2. Both sections are compulsory.

SECTION- A

1. WAP in C to add two nxn matrices and print the sum matrix. 2 marks
2. Explain the difference between structure and union in detail. 3 marks
3. WAP in C to find the factorial of positive integer number using recursive function. 3 marks

SECTION- B

1. WAP in C to sort the elements of the array using bubble sort method. 3 marks
2. Explain in details the Macro and its types with suitable example. 3 marks
3. Explain malloc(), calloc(), realloc() and free() functions used in Dynamic Memory Allocation with suitable example. 4 marks

int fib(nnn);
function

int fib(n-2);